Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/23/2007
PI Name:	Walsworth, Ronald L. Ph.D.		
Project Title:	Advanced Technology Development of Laser-Polarized Noble Gas Magnetic Resonance		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02138	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	08/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1489		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We developed laser-polarized noble gas nuclear magnetic resonance (NMR) as a novel biomedical imaging tool for ground-based and possibly space-based application. This emerging multidisciplinary technology enables high-resolution gas-space magnetic resonance imaging (MRI), e.g., of the lung, as well as studies of tissue perfusion. In addition, laser-polarized noble gases (3He and 129Xe) do not require a large magnetic field for sensitive detection, which opens the door to practical MRI at very low magnetic fields with an open-geometry device. We pursued two specific aims in the current project: (i) development of an optimized, open-geometry, low-field MRI system for human lung imaging; and (ii) development of minimally-invasive NMR technologies to probe acinar structure, gas exchange, alveolar oxygen concentration, and regional blood volume in human and animal lungs. Task objectives were: (i) design and construction of an optimized, second-generation system for open-geometry		

	low-field human MRI; and (ii) development of capabilities for MRI studies of human lungs, e.g., studies of acinar structure, gas exchange, alveolar oxygen concentration, and regional blood volume.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The techniques developed (including low-field MRI and gas-diffusion NMR of inhaled, laser-polarized noble gas) will provide novel tools for furthering understanding of cardiopulmonary physiology, and will serve as non-invasive diagnostics of, and treatment monitors for, lung diseases such as asthma and emphysema. Also, the open-geometry low-field MRI system will be applicable to patient groups for which traditional high-field MRI is prohibited or difficult: e.g., patients with implants or claustrophobia, or with size or physical constraints that prevent being placed in the tight tube of a traditional MRI instrument.
Task Progress:	We developed and tested a second-generation low-field MRI system for optimized open-access human lung imaging. The B0 coils are based on an eighth-order, bi-planar, constant-current four-coil design, providing up to 0.01 tesla with a homogeneity of 50 ppm in a 40 cm diameter spherical volume with an inter-coil distance of ~ 90 cm. The B0 coils are mounted on circular aluminum flanges that are slotted to minimize eddy currents. A water-cooled Alpha current supply provides up to 100 amperes, which run in series through all four coils with stability better than 10 ppm. The magnet is housed in an RF-shielded enclosure that provides noise attenuation better than 80 db at the 50-500 kHz range. Three planar gradient coils have been designed and constructed to provide up to 0.2 G/cm with a linearity suitable for 256 x 256 images with a 40 cm field of view. The gradient coils have greatly improved heat sinking compared to the first-generation system, and can thus generate fast (~ 1 ms) pulsed magnetic field gradients without the need for long delays between echo acquisitions. The second-generation system also features a commercial RF control and acquisition console designed to operate at kHz frequencies without modification, while a single RF transmit/receive (B1) coil reduces cross-coupling noise in a fully 50 O impedance-matched system. We investigated several options for B1 coils, and determined the most useful and practical to be a single solenoidal coil worn by the subject around the chest. As a demonstration of the second-generation low-field imager's capability, two subjects were imaged a total of 6 times, in both horizontal (supine) and vertical (sitting) body orientations. IRB approval was obtained for 3He inhalation/imaging from the Partners Human Research Committee (Massachusetts General/Brigham & Women's Hospitals) under an inter-institutional IRB agreement with the Harvard University Committee for the Use of Human Subjects. Each subject inhaled ~ 500 cm3 of laser-polarized 3He gas from a Tedlar bag. We employed 2D and 3D gradient-recalled echo sequences. 2D projection images were acquired in both orientations without slice selection over a 50 cm FOV, in ~ 5 seconds. 3D images were acquired over a 50 x 50 x 12 cm FOV, in ~ 30 seconds. The 3D imaging technique provides a clear representation of the lungs. After rendering, the pleural boundaries are clear and the cardiac cavity is clearly visible. In the 2D slices, the effect of the diaphragm is seen on the lower portion of the lungs. When the subject is sitting vertically, the characteristic curvature from the diaphragm at the bottom of the lungs seen when the subject is supine is no long visible, and the lungs are distended slightly.
Bibliography Type:	Description: (Last Updated: 07/25/2019)
Articles in Peer-reviewed Journals	Ruset IC, Tsai LL, Mair RW, Patz S, Hrovat MI, Rosen MS, Muradian I, Ng J, Topulos GP, Butler JP, Walsworth RL, Hersman FW. "A system for open-access 3He human lung imaging at very low field." Concepts in Magnetic Resonance Part B: Magnetic Resonance Engineering. 2006 Oct;29B(4): 210-21. http://dx.doi.org/10.1002/cmr.b.20075 , Oct-2006
Articles in Peer-reviewed Journals	Mair RW, Hrovat MI, Patz S, Rosen MS, Ruset IC, Topulos GP, Tsai LL, Butler JP, Hersman FW, Walsworth RL. "3He lung imaging in an open access, very-low-field human magnetic resonance imaging system." Magn Reson Med. 2005 Apr;53(4):745-9. PMID: 15799045 , Apr-2005
Articles in Peer-reviewed Journals	Wang R, Rosen MS, Candela D, Mair RW, Walsworth RL. "Study of gas-fluidization dynamics with laser-polarized 129Xe." Magn Reson Imaging. 2005 Feb;23(2):203-7. PMID: 15833613 , Feb-2005